

IA Mathematics for Natural Sciences B:

Examples Sheet 1 - Basics of vector geometry, and products of vectors

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Basics of vector geometry

1. Let $A = (1, 3, 4)$, $B = (-1, 2, 4)$ and $C = (2, 2, 3)$. Which of the vectors $\overrightarrow{AB}$, $\overrightarrow{BC}$ and $\overrightarrow{AC}$ is the longest?
2. Let $\mathbf{a} = (0, 2, 0)^T$. Describe the loci $|\mathbf{r}| = 4$, $|\mathbf{r} - \mathbf{a}| = 4$ as the vector $\mathbf{r}$ varies. What is their intersection?
3. (a) Given two vectors $\mathbf{v}$, $\mathbf{w}$, define their sum $\mathbf{v} + \mathbf{w}$. Explain why $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$.
(b) An aeroplane's engine produces a velocity 100 km h^{-1} due North. If there is a wind travelling at a velocity 80 km h^{-1} at a bearing 60° West of North, use the sine and cosine rules to determine how fast the aeroplane travels across the surface of the Earth, and its bearing from North.
4. (a) Define a *basis* of vectors.
(b) Show that $\{(1, 0)^T, (1, 1)^T\}$ is a basis for two-dimensional space. Find the components of the vector $(2, 3)^T$ in this basis.
(c) Which of the following sets of vectors are bases for three-dimensional space? Explain your reasoning in each case.
 $S_1 = \{(1, 1, 0)^T, (1, 0, 1)^T\}$, $S_2 = \{(1, 1, 0)^T, (1, 0, -1)^T, (2, 1, -1)^T\}$, $S_3 = \{(1, 0, 0)^T, (1, 1, 0)^T, (0, 0, 1)^T\}$.
5. Repeat Question 3(b) by resolving the vectors into components with respect to the basis $\{\mathbf{e}_1, \mathbf{e}_2\}$, with $\mathbf{e}_1$ due East and $\mathbf{e}_2$ due North.

Parametric vector equations

6. (a) Explain why $\mathbf{r}(\lambda) = \mathbf{v} + \lambda\mathbf{w}$, where $\mathbf{w} \neq \mathbf{0}$, is the vector equation of a line, as the parameter $\lambda \in \mathbb{R}$ varies. Why do we require $\mathbf{w} \neq \mathbf{0}$?
(b) Find, in the form of part (a), vector equations for the lines L_1 passing through $(1, 0, 1)$, $(2, 3, -1)$, and L_2 passing through $(1, 1, 0)$, $(1, -3, 4)$.
(c) By setting $\mathbf{r} = (x, y, z)^T$, and eliminating the parameter λ , find sets of Cartesian equations which describe the lines L_1, L_2 . How many Cartesian equations are needed in each case?
7. (a) Explain why $\mathbf{r}(\lambda, \mu) = \mathbf{v} + \lambda\mathbf{w} + \mu\mathbf{u}$, where $\mathbf{w}, \mathbf{u}$ are independent vectors, is the vector equation of a plane, as the parameters $\lambda, \mu \in \mathbb{R}$ vary. Why do we require $\mathbf{w}, \mathbf{u}$ independent?
(b) Find, in the form of part (a), vector equations for the planes P_1 passing through $(1, 2, 3)$, $(1, 0, 1)$, $(3, 0, -1)$, and P_2 passing through $(1, 0, 1)$, $(1, 1, 0)$, $(1, -3, 2)$.
(c) By setting $\mathbf{r} = (x, y, z)^T$, and eliminating the parameters λ, μ , find sets of Cartesian equations which describe the planes P_1, P_2 . How many Cartesian equations are needed in each case?
8. Show that in any tetrahedron, the three lines joining the midpoints of opposite edges are concurrent.

Scalar products: basics

9. (a) State two equivalent formulae for the scalar product $\mathbf{v} \cdot \mathbf{w}$ of two vectors. Under what conditions are each of these formulae valid?
(b) Draw a diagram showing the vectors $\mathbf{v} = (1, 1, 0)^T$ and $\mathbf{w} = (1, 1, 1)^T$. Calculate $\mathbf{v} \cdot \mathbf{w}$ using both formulae you stated in part (a), and check they agree. What does the sign of $\mathbf{v} \cdot \mathbf{w}$ tell you?

10. Using the two formulae for the scalar product:
- (a) determine the angles AOB and OAB , where $A = (0, 3, 4)$ and $B = (3, 2, 1)$ respectively;
 - (b) find the acute angle at which two diagonals of a cube intersect.
11. Using the properties of the scalar product, simplify $(\mathbf{v} - \mathbf{w}) \cdot (2\hat{\mathbf{v}} - \mathbf{w})$, where $\mathbf{v}, \mathbf{w}$ are perpendicular.
12. Explain why an equation for a plane passing through the point $\mathbf{a}$, normal to a vector $\mathbf{n} \neq \mathbf{0}$, is $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$. Hence, find a Cartesian equation for the plane through $(1, 0, 1)$ with normal $(0, 2, 3)^T$.
13. Prove the *Cauchy-Schwarz inequality*, $\mathbf{v} \cdot \mathbf{w} \leq |\mathbf{v}||\mathbf{w}|$ for all vectors $\mathbf{v}, \mathbf{w}$. Deduce the *triangle inequality*, $|\mathbf{v} + \mathbf{w}| \leq |\mathbf{v}| + |\mathbf{w}|$ for all vectors $\mathbf{v}, \mathbf{w}$, and give a geometric interpretation of the inequality.

Scalar products: parallel projections

14. Draw a diagram showing that the component of $\mathbf{v}$ parallel to $\mathbf{u} \neq \mathbf{0}$ is given by $\mathbf{v} \cdot \hat{\mathbf{u}}$. Using this result:
- (a) Show that an equation of a plane passing through the point $\mathbf{a}$, with normal $\mathbf{n} \neq \mathbf{0}$, is $\mathbf{r} \cdot \mathbf{n} = d$, where $d/|\mathbf{n}|$ is the shortest distance from the plane to the origin. Find the equations of the planes P_1, P_2 from Question 7 in the form $\mathbf{r} \cdot \mathbf{n} = d$, and hence, identify their normals and their shortest distances to the origin.
 - (b) Explain why an equation of the line through $\mathbf{a}$ in the direction $\mathbf{v} \neq \mathbf{0}$ is $((\mathbf{r} - \mathbf{a}) \cdot \hat{\mathbf{n}})\hat{\mathbf{n}} = \mathbf{r} - \mathbf{a}$.
15. By using the scalar product to find appropriate parallel projections, show that:
- (a) the shortest distance from a point $\mathbf{p}$ to a plane passing through a point $\mathbf{a}$ with normal $\mathbf{n}$ is given by $|(\mathbf{p} - \mathbf{a}) \cdot \hat{\mathbf{n}}|$, and the closest point on the plane to the point is $\mathbf{p} - [(\mathbf{p} - \mathbf{a}) \cdot \hat{\mathbf{n}}]\hat{\mathbf{n}}$;
 - (b) the shortest distance from a point $\mathbf{p}$ to a line passing through a point $\mathbf{a}$ with direction vector $\mathbf{v}$ is given by $\sqrt{|\mathbf{p} - \mathbf{a}|^2 - |(\mathbf{p} - \mathbf{a}) \cdot \hat{\mathbf{v}}|^2}$. Does this help us find the closest point on the line?
- Hence, use the scalar product to find the shortest distance from the points $P = (1, 0, 1), Q = (1, 0, -1)$ to P_1, P_2 and L_1, L_2 from Questions 6 and 7. Do P, Q lie on the same side of the plane P_1 ?
16. (a) Describe the surface given by the vector equation $|\mathbf{r} - (\mathbf{r} \cdot \mathbf{u})\mathbf{u}| = 2$, where $\mathbf{u} = (1, 0, 1)^T / \sqrt{2}$. What is the intersection of this surface with the surface $x + z = 0$?
- (b) Describe the surface with vector equation $\mathbf{r} \cdot \mathbf{u} = m|\mathbf{r}|$, where $\mathbf{u} = (1, 0, 1)^T / \sqrt{2}$ and $|m| < 1$.

Vector products: basics

17. (a) State two equivalent formulae for the vector product $\mathbf{v} \times \mathbf{w}$ of two vectors. Under what conditions are each of these formulae valid?
- (b) Draw a diagram showing the vectors $\mathbf{v} = (1, 1, 0)^T$ and $\mathbf{w} = (1, 1, 1)^T$. Calculate $\mathbf{v} \times \mathbf{w}$ using both formulae you stated in part (a), and check they agree. Can we determine the angle between $\mathbf{v}, \mathbf{w}$ using the vector product?
18. Using the vector product:
- (a) find an equation of the form $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$ for the plane through $(1, 1, 1), (1, 2, 3)$ and $(0, 0, 4)$;
 - (b) determine whether the points $(0, 0, 2), (0, 1, 3), (1, 2, 3), (2, 3, 4)$ are coplanar.
19. Using the properties of the vector product, show that $(\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a}) = \mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}$.
20. Explain why an equation for a line passing through the point $\mathbf{a}$, in the direction $\mathbf{v} \neq \mathbf{0}$, is $(\mathbf{r} - \mathbf{a}) \times \mathbf{v} = \mathbf{0}$. Hence, find Cartesian equations for the line through the points $(1, 0, 1), (0, 2, 3)$.
21. Derive a formula for the shortest distance between the lines $\mathbf{r}_1(\lambda) = \mathbf{a}_1 + \lambda\mathbf{v}_1$ and $\mathbf{r}_2(\mu) = \mathbf{a}_2 + \mu\mathbf{v}_2$. What happens if $\mathbf{v}_1 \times \mathbf{v}_2 = \mathbf{0}$? Find the shortest distance between the lines L_1, L_2 in Question 6.